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SUPERSEDING
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15 November 1966

MILITARY SPECIFICATION

MAGNET MATERIALS, PERMANENT

This specification is approved for use by all activities of the Department of the Army.

1. SCOPE

1.1 Scope. This specification covers permanent magnet materials in the form of bars, rods, strips, wire, castings, and sintered shapes.

1.2 Classification. The permanent magnet materials shall be of the following classes and forms, as specified (See 6.2):

Classes 1/	Forms
A1	Wrought bars and rods
A2	Wrought bars, rods, and strips
A3	Wrought bars and rods; castings
B1, B2, B3, B4, B7, B8, B10	Castings
B5, B6, B9, B11	Sintered shapes
C1	Wrought bars, rods, strips, and wire; castings
C2	Wrought bars, rods, strips, and wire
C3	Sintered shapes
C4	Wrought bars, rods, strips, and wire

1/ Nominal chemical compositions for the various classes are given in Table VI (see 6.6).

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-163	Steel Mill Products Preparation for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Scientific Papers No. 361 and No. 383

Journal of Research, Volume 23 (1939)

(Application for copies should be addressed to the National Bureau of Standards, Washington, DC, 20013.)

Air Force Manual (AFM-71-4)

TM 38-250	Packaging and Handling of Dangerous Materials for Transportation by Military Aircraft.

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material. Raw materials used shall be such as to produce permanent magnet materials meeting all requirements of this specification.

3.2 Condition.

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3.2.1 Class A. Unless otherwise specified (see 6.2), Class A wrought material shall be furnished in the as-rolled, as-forged, or as-drawn condition, as applicable, and Class A3 cast material in the as-cast condition.

3.3.3 Class B and class C. Unless otherwise specified (see 6.2), Class B and Class C wrought, cast, or sintered material shall be furnished in the fully heat-treated condition and shall be completely demagnetized.

3.3 Heat-treatment. The material shall be given such uniform treatment as the manufacturer desires to produce material which will conform to the requirements specified, applied in a manner that will produce the utmost uniformity. Heat-treating operations shall be performed on the whole of a piece, never on a part only.

3.4 Stabilization. Where stabilization treatment is required, material shall be stabilized in accordance with the requirements specified in the contract or purchase order (see 6.2).

3.5 Magnetic properties. Material covered by the specification, after suitable heat-treatment, shall have the coercive forces and residual indictions specified in Table I when determined from the corresponding indicated values of peak magnetizing force specified.

3.6 Dimensions and tolerances.

3.6.1 Bars and rods.

3.6.1.1 Diameter, thickness, and width. The diameter, thickness, and width of bars and rods shall not vary from that ordered by more than the amount specified in Table II.

3.6.1.2 Length. Unless otherwise specified (see 6.2), bars and rods shall be furnished in random lengths of not less than 6 feet.

3.6.2 Strips. The thickness and width of strips shall not vary from that ordered by more than the amounts specified in Table III.

3.6.3 Wire. Diameter or thickness of wire, both round and rectangular, shall not vary from that ordered by more than the permissible variation specified in Table IV.

3.6.4 Castings and sintered shapes. The dimensions of castings and sintered shapes shall be as specified on the drawings, within the tolerances given thereon. Sufficient stock shall be allowed for finishing where indicated by the drawings.

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3.7 Workmanship.

3.7.1 Bars and rods. Bars and rods shall be commercially straight and sound, uniform in quality and condition, free from pipes, laps, cracks, twists, deep rolling marks, damaged ends, excessive non-metallic inclusions, segregation, seams, scale, and other injurious defects.

3.7.2 Strips. Strips shall be free from seams, laminations, blisters, excessive or loose scale, and other injurious defects.

3.7.3 Wire. Wire shall be sound, uniform in quality and condition, free from kinks, twists, deep die scratches, and other injurious defects.

3.7.4 Castings. Castings shall be sound, clean, uniform in quality and condition, and free from injurious shrinkage defects, cracks, blowholes, hard spots, sand, excessive porosity, non-metallic inclusions, and other injurious defects which make the castings unsuitable for use in the intended application.

3.7.5 Sintered shapes. Sintered shapes shall be sound, uniform in quality and condition, and free from cracks, hard spots, excessive porosity, and other injurious defects.

3.7.6 Cutting and punching. Bars, rods, and strips shall be in such condition that they may be cut or punched into blanks satisfactorily by the method stated in the contract or purchase order (see 6.2).

3.7.7 Cracking. Class A material shall not crack when properly heat treated by the purchaser.

3.7.8 Welding. Welding shall not be permitted, except when specifically authorized in the contract or purchase order (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Sampling, inspection, and test methods.

4.2.1 Sampling.

4.2.1.1 Lot. A lot shall comprise all material, or finished magnets, of the same class, the same condition, the same form, and the same size submitted for inspection at one time. In addition, a lot of Class A material shall be from but one heat.

4.2.2 Specimens for magnetic test.

4.2.2.1 Number of specimens.

4.2.2.1.1 Wrought material. At least 2 specimens for magnetic test shall be selected for each 2,000 pounds of Class A1, A2, or A3 material, or for each 20 pounds of Class C1, C2, or C4 material, or fraction thereof, in each lot, except that not more than five specimens shall be taken from any lot.

4.2.2.1.2 Cast or sintered material or finished magnets. The number of specimens to be selected from each lot of Class A3, B1, B2, B3, B4, B7, B8, B10, or C1 castings, or Class B5, B6, B9, B11, or C3 sintered shapes, or of finished magnets of any class shall be determined from Table V. If the number of defective specimens in the sample does not exceed the acceptance number, the entire lot is acceptable. If number of defective specimens in the sample is equal to or greater than the rejection number, the entire lot is rejectable.

4.2.2.2 Form of specimens. Specimens for magnetic tests shall be of the form best suited for use with the magnetic test apparatus to be employed. When special tests on finished magnets have been approved, in lieu of test of the magnet material, the specimens shall be the actual magnets being produced.

4.2.2.3 Heat-treatment and aging.

4.2.2.3.1 Class A material. Specimens for magnetic test shall be given the heat-treatment, including aging, which is to be used in the production of finished magnets from the lot which the specimens represent. Before testing, Class A1 and Class A2 material shall be aged at room temperature for at least 3 days, and Class A3 material for at least 10 days, after heat-treatment.

4.2.2.3.2 Class B and class C material. Specimens for magnetic test shall be heat-treated with the actual pieces of the lot which the specimens represent. The specimens need not be aged before testing.

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4.2.3 Methods of magnetic test.

4.2.3.1 Tests of magnetic material. Magnetic tests shall be made by one of the methods indicated below as suitable for the class of material to be tested.

4.2.3.1.1 Isthmus method. The isthmus Method, as described in National Bureau of Standards Scientific Papers No. 361 and No. 383, may be used for material of all classes.

4.2.3.1.2 High-H permeameter. The High-H Permeameter, as described in National Bureau of Standards Journal of Research, Volume 23 (1939), may be used for material of all classes, except Class C4.

4.2.3.1.3 Simplex Super-H adapter. The Simplex Super-H Adapter may be used for material of Classes A1, A2, A3, B1, B2, B5, and C2.

4.2.3.1.4 Saturation permeameter. The Saturation Permeameter may be used for material of Classes A3, B1, B2, B3, B4, B5, B6, B7, B8, B9, B10, B11, C1, C2, and C3.

4.2.3.1.5 Babbitt permeameter. The Babbitt (or J) Permeameter may be used for material of Classes A1, A2, and A3.

4.2.3.1.6 Burrows compensated double yoke. The Burrows Compensated double yoke may be used for material of Classes A1, and A2.

4.2.3.2 Alternate test on finished magnets. When specified on drawings or when approved by the contracting officer, a test on finished magnets may be used in lieu of the tests of magnetic material prescribed above.

4.3 Cracking of Class A material during heat-treatment by purchaser.

4.3.1 3 percent or less cracked. If not more than 3 percent of a shipment cracks during heat-treatment, the shipment shall be considered acceptable insofar as the requirement for resistance to cracking in concerned.

4.3.2 More than 3 percent, less than 6 percent cracked. If more than 3 percent, but not more than 6 percent, of a shipment cracks during heat-treatment, those pieces which crack shall be subject to rejection, but the remainder of the shipment shall be considered acceptable insofar as the requirement for resistance to cracking is concerned.

4.3.3 More than 6 percent cracked. If more than 6 percent of a shipment cracks during heat-treatment, the entire shipment shall be subject to rejection. At the option of the purchaser, an entire shipment may be rejected, if, when 20 percent or more of a shipment has been heat-treated, more than 6 percent of the portion heat-treated has cracked.

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option of the purchaser, an entire shipment may be rejected. if, when 20 percent or more of a shipment has been heat-treated, more than 6 percent of the portion heat-treated has cracked.

4.4 Inspection. All material shall be inspected for workmanship, and a sufficient number of measurements taken to insure compliance with dimensional requirements.

4.5 Rejection. Failure of any specimen to conform to any one or more of the requirements of this specification shall be cause for rejection of the lot, except where a sampling plan is provided.

4.6 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be examined for compliance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified (see 6.2), preservation, packaging, packing, and marking shall be as specified herein.

5.2 Packaging and packing. Packaging and packing shall be level A or C, as specified (see 6.2), in accordance with MIL-STD-163.

5.3 Marking. In addition to any special marking required by the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129 and TM 38-250, 11-19 (AFM-71-4).

6. NOTES

6.1 Intended use. The material covered by this specification is intended for use in the manufacture of permanent magnets.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Class and form of material required (see 1.2).

c. Condition of Class A wrought material, if other than specified (see 3.2.1).

d. Condition of Class B and Class C wrought, cast, or sintered material, if other than specified (see 3.2.2).

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e. Requirements for stabilization treatment, if required (see 3.4).

f. Length of bars and rods, if other than specified (see 3.6.1.2).

g. Method to be used for cutting or punching bars, rods, and strips into blanks (see 3.7.6).

h. Whether welding is permitted (see 3.7.8).

i. Whether castings or sintered shapes are to be machined, ground, or magnetized.

j. Requirements for preservation, packaging, packing, and marking, if different (see 5.1).

k. Applicable level(s) of packaging and packing (see 5.2).

l. Special marking requirements, if any (see 5.3).

6.3 Design information.

6.3.1 Forms.

6.3.1.1 Bars, rods, and strips. Class A bars, rods, and strips should be used where low or moderate magnetic propertics are sufficient, or where forgeability is essential. Higher magnetic properties are obtainable in Class C1, C2, and C4 wrought bars, rods, strips, and wire. All classes of bars and rods covered by this specification can be machined under proper conditions.

6.3.1.2 Castings. Castings should be used where high magnetic properties are required, except for very small (under 0.1 pound) or intricate pieces in which case sintered shapes are generally to be preferred. Except for Classes A3 and C1, castings are not machinable and must be finished by grinding, unless cast with soft steel inserts.

6.3.1.3 Sintered shapes. Sintered shapes should be used where high magnetic properties are required in small or intricate pieces. For pieces larger than 0.1 pound, castings are generally to be preferred. Sintered shapes are not machinable after final sintering and must be finished by grinding. Except when large quantities (over 100,000) of a shape are required, commercial shapes should be used when practicable. The tensile strength of sintered shapes is considerable greater than that of the cast forms.

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6.3.2 Classes.

6.3.2.1 Class A1 alloy. A low-cost bar stock material suitable for use where relatively low magnetic properties are required and where permanence of magnetic properties is not essential. It is forgeable and machinable, and requires water-quenching.

6.3.2.2 Class A2 alloy. A widely-used class of bar stock for magnets of moderate magnetic properties. It is forgeable and machinable, and is oil-hardening.

6.3.2.3 Class A3 alloy bar stock. Should be used only where forgeability or machinability, together with magnetic properties higher than those of Class A2, is essential. Class B1 castings, where their use is feasible, are generally to be preferred to Class A3 alloy due to better magnetic properties and lower cost.

6.3.2.4 Class A3 castings. Should be used only when machinable castings are necessary.

6.3.2.5 Class B1 castings. Widely used where economical high magnetic properties are desired. Class B1 alloy cannot be forged or machined, and must be finished by grinding.

6.3.2.6 Class B2 castings. Should be used where high magnetic properties, not obtainable with Class B1, are necessary. Class B2 cannot be forged or machined, and must be finished by grinding.

6.3.2.7 Class B3 castings. Should be used only when a relatively high coercive force is necessary. This may be the case when the magnets must be very short. Class B3 alloy cannot be forged or machined, and must be finished by grinding.

6.3.2.8 Class B4 castings. Should be used where the highest available combination of magnetic properties is necessary. Class B4 alloy cannot be forged or machined, and must be finished by grinding.

6.3.2.9 Class B5 sintered shapes. Should be used in place of Class B2 castings when consideration of size or shape indicates the use of sintered shapes.

6.3.2.10 Class B6 sintered shapes. Should be used only where a relatively high coercive force is necessary and where considerations of shape and size indicate the use of sintered shapes.

6.3.2.11 Class B7 castings. Have a high coercive force and a slightly lower residual induction and energy product than Class B4

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castings. Class B7 castings cannot be forged or machined and must be finished by grinding.

6.3.2.12 Class B8 castings. Have a high coercive force than Class B1, B2, B3, B4, and B7 alloys. Class B8 castings cannot be forged or machined, and must be finished by grinding.

6.3.2.13 Class B9 sintered shapes. Should be used instead of Class B4 castings when considerations of shape and size indicate the use of sintered shapes.

6.3.2.14 Class B10 castings. Have the highest coercive force of all Class B alloys. Class B10 castings cannot be forged or machined, and must be finished by grinding.

6.3.2.15 Class B11 sintered shapes. Have a higher coercive force than Class B5, B6, and B9 sintered shapes, and should be utilized only when a high coercive force is needed and where considerations of size and shape indicates the use of sintered shapes.

6.3.2.16 Class C1 alloy. Used when castings requiring special machining are desired. It may be obtained also as bars, rods, wire, and as punchings from strip. It is ductile and machinable prior to final heat-treatment and has magnetic properties slightly inferior to Class C2 alloy.

6.3.2.17 Class C2 alloy. A ductile wrought material with moderate magnetic properties and is used when ductility after heat-treatment is desired or when parts are to be formed from wire or finished from strip.

6.3.2.18 Class C3 sintered shapes. Have a high coercive force, very high resistivity, and low density. It may be finished by grinding, but it is very weak physically and must be handled with care.

6.3.2.19 Class C4 alloy. Used for small magnets under open-circuit conditions. This material is very expensive. It can be cold worked and is usually used in the form of punchings or machined parts.

6.4 Aging. Following the heat-treatment, Class A alloys undergo a slow metallurgical change at room temperature and lose a portion of their field strength. By proper treatment, this change can be artifically accelerated so that the change does not occur after testing or during service. Loss of field strength through aging is very slight or completely absent in Class B and Class C materials.

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6.5 Stabalization. The field strength of permanent magnet materials is subject to change due to the influence of mechanical shock or vibration, external magnetic fields, and temperature changes not involving metallurgical changes. The effects of these influences can be minimized by certain stabilization treatments. Magnet alloys of Classes B and C require comparatively little stabilization and in many applications none is required.

6.6 Nominal compositions of the alloys. Nominal compositions covered by this specification are shown in Table VI. The compositions are given for information only and not as a basis for acceptance of material. Modifications of these compositions are used by various manufacturers to meet the specified magnetic properties.

6.7 Supersession data. This specification includes the requirements of Missile Interim Specifications MIS-13123, dated 15 November 1966.

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Table I. Magnetic properties

Class	Coercive force (H_c) minimum	Residual induction (B_r) minimum	Peak magnetizing force (II_{max})
	Oersteds	Gausses	Oersteds
A1	40	8700	300
A2	62	9000	300
A3	210	9000	1000
B1 1/	450	6700	2000
B1 2/	400	7100	2000
B2	540	7200	2000
B3	700	5200	3000
B4	575	12000	3000
B5	520	6900	2000
B6	700	5200	3000
B7	750	10000	3000
B8	950	5800	3000
B9	575	10000	3000
B10	1470	8625	5000
B11	1400	7500	5000
C1	660	3400	3000
C2 3/	550	5400	2000
C2 4/	440	4800	2000
C3	1000	1600	3500
C4	500 5/	550	20000

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Table I. Magnetic properties
(Continued)

NOTES:

1/ Up to 5/8 by 5/8-inch cross-sectional dimension, inclusive.

2/ Over 5/8 by 5/8-inch cross-sectional dimension.

3/ Up to 0.155 inch diameter, inclusive.

4/ Over 0.155 inch diameter.

5/ The minimum required instrinsic coercive force is 6000 oersteds.

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Table II. Permissible variations in diameter, thickness, and width of bars and rods.

ROUNDS, SQUARES, AND ROUND-CORNERED SQUARES

Ordered diameter or distance between parallel faces		Permissible variation		
Over	To and including	Plus	Minus	Out-of-section 1/
Inches	Inches	Inch	Inch	Inch
	5/16	0.005	0.005	0.008
5/16	7/16	0.006	0.006	0.009
7/16	5/8	0.007	0.007	0.010
5/8	7/8	0.008	0.008	0.012
7/8	1	0.009	0.009	0.013
1	1-1/8	0.010	0.010	0.015
1-1/8	1-1/4	0.011	0.011	0.016
1-1/4	1-3/8	0.012	0.012	0.018
1-3/8	1-1/2	0.014	0.014	0.021
1-1/2	2	1/64	1/64	0.023
2	2-1/2	1/32	0	0.023
2-1/2	3-1/2	3/64	0	0.035
3-1/2	4-1/2	1/16	0	0.046
4-1/2	5-1/2	5/64	0	0.058
5-1/2	6-1/2	1/8	0	0.070
6-1/2	8	5/32	0	0.085

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Table II. Permissible variations in diameter, thickness, and width of bars and rods (Continued)

HEXAGONS AND OCTAGONS				
Ordered distance between parallel faces		Permissible variation		
Over	To and including	Plus	Minus	Out-of-section 1/
Inches	Inches	Inch	Inch	Inch
	1/2	0.007	0.007	0.011
1/2	1	0.010	0.010	0.015
1	1-1/2	0.021	0.013	0.025
1-1/2	2	1/32	1/64	1/33
2	2-1/2	3/64	1/64	3/64
2-1/2	3-1/2	1/16	1/64	1/16

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Table II. Permissible variations in diameter, thickness, and width of bars and rods
(Continued)

FLATS (SQUARE-EDGE AND ROUND-EDGE)								
Width ordered		Permissible variation in width		Permissible variation in thickness, plus or minus				
Over	To and including	Plus	Minus	Thickness Under 1/4 inch	Thickness 1/4 to 1/2 inch, inclusive	Thickness Over 1/2 to 1 inch inclusive	Thickness Over 1 to 2 inches, inclusive	Thickness Over 2 to 4 inches, inclusive
Inches	Inches	Inch	Inch	Inch	Inch	Inch	Inch	Inch
	1	1/64	1/64	0.007	0.008	0.010		
1	2	1/32	1/32	0.007	0.012	0.015	1/32	
2	4	1/16	1/32	0.008	0.015	0.020	1/32	3/64
4	6	3/32	1/16	0.009	0.015	0.020	1/32	1/16

NOTE:

1/ Out-of-round is the difference between the maximum and minimum diameters of the bar, measured at the same cross-section. Out-of-square-section, out-of-hexagonal-section, and out-of-octagonal-section are the differences in the two, three, or four dimensions at the same cross-section of the bar, each dimension being the distance between parallel faces.

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Table III. Permissible variation in strip dimensions.

Thickness ordered		Permissible variation in thickness, plus or minus 1/			
Over	To and including	Width to 3-1/2 inches, inclusive	Width over 3-1/2 in. to 6 in., inclusive	Width over 6 in. to 12 in., inclusive	Width over 12 in. to 15 in., inclusive
Inch	Inch	Inch	Inch	Inch	Inch
0.0636	0.1120	0.004	0.005	0.005	0.006
0.1120	0.1718	0.005	0.005	0.006	0.007
0.1719	0.2030	0.006	0.006	0.006	0.007
0.2031	0.2499	0.006	0.006	0.007	0.008

Width ordered		Permissible variation in width, plus or minus
Over	To and including	
Inches	Inches	Inch
	2	1/32
2	6	3/64
6	12	1/16
12	15	3/32

1/ Thickness is measured at any point not less than 3/8 inch from the edge of the strip.

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Table III. Permissible variation in strip dimensions
(Continued)

NOTE:

1/ For thickness measurements taken 3/8-inch from edge of strip. Measurements taken at the center of strip may be larger than those taken 3/8-inch from edge by the following amounts:

a.	For widths up to inches, inclusive:	0.002 inch
b.	For width over 2 inches to 5 inches, inclusive:	0.003 inch
c.	For widths over 5 inches to 10 inches, inclusive:	0.004 inch
d.	For widths over 10 inches to 15 inches, inclusive:	0.005 inch

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Table IV. Permissible variations in wire dimensions

ROUND WIRE	
Diameter ordered	Permissible variation in diameter, plus or minus
Inches	Inch
To 0.150, inclusive	0.002
Over 0.150 to 0.250, inclusive	0.004
RECTANGULAR WIRE	
Width ordered	Permissible variation in Width, plus or minus
To 1-1/4, inclusive	0.005
Thickness ordered	Permissible variation in thickness, plus or minus
To 0.150, inclusive	0.002
Over 0.150 to 0.250, inclusive	0.004

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Table V. Sampling plan for magnetic test of cast or sintered magnet materials and finished magnets

Lot size	Sample size	Acceptance number	Rejection number
Under 25	10	1	2
26 through 50	15	1	2
51 through 100	20	1	2
101 through 200	30	2	3
201 through 300	40	3	4
301 through 500	55	3	4
501 through 800	75	4	5
801 through 1300	115	6	7
1301 through 3200	150	8	9

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Table VI. Nominal chemical composition, in percent.

Class	Commercial Designation	C	Mn	Si	Al	Co	Cr	Cu	Ni	W	Ti	Fe
A1	Mn Steel	0.85	0.70	0.15								Remainder
A2	Cr steel	0.90	0.45	0.15			3.5					Remainder
A3	Fe-Co alloy	0.75	0.50	0.15		36	3.5			4		Remainder
B1	Alnico 3, cast	0.10	0.15	0.15	12				25			Remainder
B2	Alnico 2, cast	0.10	0.15	0.15	10	12.5		6	17			Remainder
B3	Alnico 4, cast	0.10	0.15	0.15	12	5			28			Remainder
B4	Alnico 5, cast	0.10	0.15	0.15	8	24		3.25	14			Remainder
B5	Alnico 2, sintered	0.10	0.15	0.15	11	12.5		6	16			Remainder
B6	Alnico 4, sintered	0.10	0.15	0.15	12	5			28			Remainder
B7	Alnico 6, cast	0.10	0.15	0.15	8	24		3	15		1.25	Remainder
B8	Alnico 12, cast	0.10	0.15	0.15	6	35			18		8	Remainder
B9	Alnico 5, sintered	0.10	0.15	0.15	8	24		3.25	14			Remainder
B10	Alnico 8, cast				7	35		4	15		5	Remainder
B11	Alnico 8, sintered				7.5	34		3.5	14		5	Remainder

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Table VI. Nominal chemical compostion, in percent
(Continued)

Class	Commerercial Designation	C	Mn	Si	Al	Co	Cr	Cu	Ni	W	Ti	Fe
C1	Cunico					29		50	21			
C2	Cunife							60	20			20
C3	Vectolite	30 Fe_2O_3, 26 Co_2O_3, 44 Fe_3O_4.										
C4	Silmanal	86.75 Ag, 8.8 Mn, 4.45 Al.										

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